

Exercise 3 - CH₂FCI bond dissociation energy

1) Total energy comparison, three methods

Method	Basis	E _{total} / Hartree
HF	STO-3G	-591.172393
B3LYP	cc-pVDZ	-599.362053
CCSD	cc-pVTZ	-598.670494

2) Electron density and electrostatic potential map (B3LYP/cc-pVDZ)

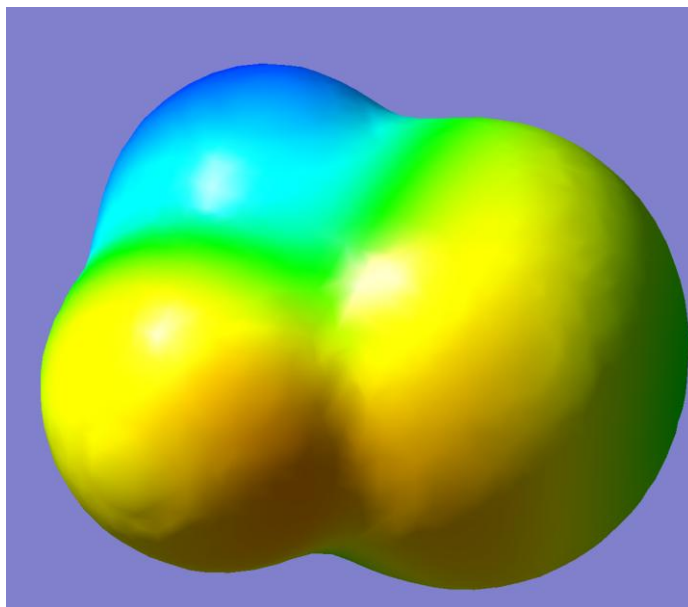


Figure 1. Total electron density surface (isovalue 0.0004) of CH₂FCI, mapped with the electrostatic potential (GaussView, B3LYP/cc-pVDZ). Blue = more positive potential, yellow/orange = more negative potential.

The surface itself is a total electron density isosurface (isovalue 0.0004), so its shape traces the outer envelope of the electron cloud: it bulges out furthest around Cl, which has the most electrons and the most diffuse (largest) electron cloud of the four elements present, and is noticeably tighter around the H atoms, which contribute only a single 1s electron each. Mapping the electrostatic potential onto this surface then shows where that density sits relative to the nuclear charges: the H atoms (blue) carry the most positive potential, while F and Cl (yellow/orange) are more negative, since both pull electron density toward themselves more strongly than C or H.

3) C-H and C-F bond dissociation energies

Strategy: homolytic bond cleavage; each radical/atom fragment is separately optimised (with frequencies, for ZPE) at UB3LYP/cc-pVDZ.

$$\text{BDE} = [\text{E}(\text{radical}) + \text{ZPE}(\text{radical}) + \text{E}(\text{atom})] - [\text{E}(\text{parent}) + \text{ZPE}(\text{parent})]$$

C-H bond, B3LYP/cc-pVDZ: CH₂FCl -> CHFCl. + H.

Species	E / Hartree	ZPE / Hartree
CH ₂ FCl (parent)	-599.362053	0.030710
CHFCl. radical	-598.698761	0.017241
H atom	-0.501258	0 (atom, no vibration)

BDE(C-H) = 0.148565 Hartree = 390.1 kJ/mol

C-F bond, B3LYP/cc-pVDZ: CH₂FCl -> CH₂Cl. + F.

Species	E / Hartree	ZPE / Hartree
CH ₂ FCl (parent)	-599.362053	0.030710
CH ₂ Cl. radical	-499.461337	0.022429
F atom	-99.726601	0 (atom, no vibration)

BDE(C-F) = 0.165834 Hartree = 435.2 kJ/mol

The C-F bond is the stronger of the two (435.2 vs 390.1 kJ/mol); the H atom is the easier of the two to remove from CH₂FCl.